Dual-Prompt-Driven Cross-Modal Fusion Learning for Few-Shot Hyperspectral Image Classification

Yuhang Li, Jinrong He, *Member, IEEE*, Xiangqing Zhang, and Zhaokui Li, *Member, IEEE*

***Abstract*—Vision-language multimodal learning has exhibited remarkable advantages in few-shot hyperspectral image (HSI) classification, where prompt learning effectively enhances feature-extraction accuracy and representation quality by guiding the model to focus on critical information. However, static prompts lack flexibility, while dynamic prompts suffer from unstable generation. To address these issues, this article proposes a dual-prompt-driven cross-modal fusion learning (DPCFL) method. Specifically, we utilize static prompt templates to provide stable prior guidance for the model. Simultaneously, a unique learnable prompt vector is designed for each category, which operates independently of the pretrained model's input to effectively mitigate potential influences from prior semantics. To alleviate semantic bias, we further design a multiloss joint optimization strategy incorporating parameter space, feature space, and cross-modal space consistency constraints, thereby improving the robustness of class-level prototype features. In addition, to tackle data scarcity, a cross-domain collaborative training mechanism is introduced to facilitate knowledge transfer. The experimental results on multiple standard HSI datasets confirm its superior classification performance under both few-shot and cross-domain scenarios. The related code will be made publicly available at the following URL: https://github.com/AIYAU/DPCFL**

***Index Terms*—Cross-modal, few-shot learning, fusion learning, hyperspectral image (HSI) classification, prompt learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide rich spectral information beyond conventional RGB images, offering unique advantages in precision agriculture, urban planning, and geological exploration [1]. The integration of their continuous spectral bands with spatial information transcends the expressive limits of traditional imagery, enabling effective acquisition and analysis of subtle material differences and intrinsic properties. However, the high-dimensional nature of the spectral data introduces redundancy and complicates processing and analysis [2], making accurate HSI classification both crucial and challenging. Early studies relied solely on hand-crafted features, followed by classifiers such as support-vector machines (SVMs) [3] and K-nearest neighbors (KNNs) [4]. These approaches are prone to subjective errors and, by focusing exclusively on spectral information while neglecting spatial context, discard critical contextual and structural relationships. Consequently, Shen and Jia [5] incorporated spatial information and, from a spectral–spatial fusion perspective, further improved classification performance.

To mitigate subjective errors, deep-learning methods have been introduced for their powerful self-learning capabilities. The pioneering spectral–spatial framework by Li et al. [6] overcame the limitations of traditional approaches, inspiring subsequent work on spectral–spatial fusion algorithms and architectures. Chen et al. [7] designed a contextual deep-learning algorithm that explicitly fuses spectral and spatial features with supervised fine-tuning. The introduction of 3-D CNNs [8] and spectral–spatial residual networks [9] further enabled deeper exploitation of spatiotemporal correlations in hyperspectral data. Huang et al. [10] introduced multiinstance factor analysis into superpixel representation, preserving both local and global information through multi-scale superpixels and decision fusion.

Despite these advances, deep models still rely on large amounts of labeled data, which is costly and time-consuming to obtain. Few-shot learning has thus emerged as a key direction to address this limitation [11]. Data augmentation [12], meta-learning [13], and domain transfer learning [14], [15], [16], [17] have been successfully explored under data-scarce conditions. However, these methods rely solely on the visual modality and struggle to capture intrinsic class characteristics in complex scenes, leading to insufficient feature discriminability. Multimodal methods, such as CLIP [18], have proven capable of encoding semantics beyond visual information. Consequently, multimodal learning has been introduced into HSI classification to enhance discriminability. Li et al. [19] used one-hot encoding with a prototype generator, while Li et al. [20] adopted cross-modal alignment with masked augmentation.

However, textual information introduces inherent semantic bias that can negatively affect model performance. Prompt learning has been proposed as an effective solution in large language models (LLMs) to improve textual feature extraction and cross-modal alignment [21]. In natural images, prompt learning has shown success; for example, Zhou et al. [22] introduced trainable text tokens to dynamically adjust textual representations. Yet applying prompt learning to HSI remains

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Yuhang Li, Jinrong He, and Xiangqing Zhang are with the College of Mathematics and Computer Science, Yan'an University, Yan'an 716000, China (e-mail: liyuhang@yau.edu.cn; hejinrong@yau.edu.cn; zhangxiangqing@yau.edu.cn).

Zhaokui Li is with the School of Computer Science, Shenyang Aerospace University, Shenyang 110136, China (e-mail: lzk@sau.edu.cn).

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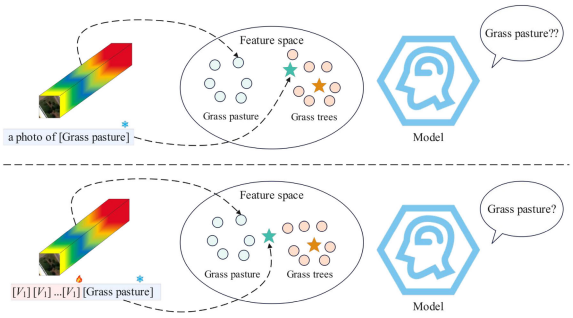


Fig. 1. Effects of different prompting strategies in HSI. The upper part is the hard prompting strategy. The lower part is the soft prompting strategy.

challenging due to significant semantic bias, and existing methods cannot effectively guide HSI features toward their respective class centers. As illustrated in Fig. 1, using existing hard-prompt (top of Fig. 1) and soft-prompt (bottom of Fig. 1) strategies, the model may fail to effectively distinguish between similar categories.

To address these issues, we propose a dual-prompt-driven cross-modal fusion learning (DPCFL) method, built upon the prototypical network framework, which integrates the strengths of hard and soft prompts to counteract textual semantic bias. Specifically, we design static prompt templates tailored to HSI to provide stable prior guidance. In addition, we depart from conventional soft-prompt learning by assigning each class a distinct learnable prompt vector decoupled from the pretrained model inputs, avoiding interference from prior semantics. To ensure stable class representation, we introduce a hierarchical alignment constraint mechanism that regulates optimization across parameter, feature, and cross-modal spaces, preventing semantic drift. The main contributions of this article can be summarized as follows.

1) To address the widespread issue of textual semantic bias in HSI, we propose the DPCFL method, which combines the advantages of hard prompts and soft prompts to improve the model's semantic alignment capability and performance in HSI classification tasks.
2) To provide stable and independent prior guidance for HSI tasks, we propose a static prompt template suitable for HSI. In addition, to avoid potential interference from the pretrained model's prior semantics, we break through the inherent design paradigm of soft prompt learning, enabling it to operate independently of pretrained model inputs, thereby achieving more flexible and stable category representation.
3) To ensure that learnable prompt vectors can stably represent their corresponding categories and prevent semantic deviation, we propose a multilevel alignment constraint mechanism that imposes constraints and optimizations across three dimensions: parameter space, feature space, and multimodal space, ensuring semantic consistency throughout the learning process.

The remaining structure of this article is organized as follows. Section II reviews the relevant background knowledge of the proposed method. Section III details the core DPCFL method. Section IV presents the experimental design and implementation. Section V summarizes the research findings and discusses future research directions.

II. Related Works

A. Few Shot Learning

Few-shot learning is a technical solution proposed to address data scarcity issues, where the number of samples available for model training is extremely limited, typically with no more than ten labeled samples per category [23]. Current research primarily employs methods such as data augmentation, metric learning, model fine-tuning, and meta-learning to enhance model generalization under low-sample conditions. Specifically, data augmentation increases data diversity by generating sample variants [24]. Metric learning aims to learn an effective feature space where samples from the same class are closer while those from different classes are farther apart [25]. Model fine-tuning leverages knowledge from models pretrained on large-scale datasets and adapts them with a small number of new samples [26]. Meta-learning, on the other hand, focuses more on "learning how to learn," training models to quickly adapt to new tasks. The training process often adopts an episode training approach, where the model learns rapid adaptation by simulating a series of small tasks with limited samples. Each task is typically structured in an N-way-K-shot-Q-query format [27] to simulate sample-constrained scenarios.

B. Multimodal Learning

Multimodal learning focuses on integrating data from different perceptual channels (such as vision, audition, and text) to construct more comprehensive and robust representations of the real world by fusing complementary intermodal information [28]. The core research in this field lies in designing effective feature alignment and fusion mechanisms to overcome modal heterogeneity and explore cross-modal semantic correlations, thereby enhancing model performance in multimodal understanding, reasoning, and generation tasks. Due to the complexity of HSI annotation, multimodal learning has also seen application expansion and innovative breakthroughs in this field. Some researchers extract semantic information from label data to guide feature extraction and classification of HSIs [19], [20]. Li et al. [29] employed a collaborative metric strategy at the decision-making stage to enhance classification performance under the guidance of textual information. In addition, Feng et al. [30] achieved efficient integration of cross-modal information by combining HSIs with synthetic aperture radar (SAR) data using contrastive learning methods. Meanwhile, Yu et al. [31] effectively preserved spatial structures and reduced spectral redundancy through multisemantic spatial attention and spectral correlation self-attention modules, significantly improving the performance of HSI classification. Recent semantic-guided vision-language works in remote sensing offer complementary insights. Sun et al. [41] leverage semantic embeddings for few-shot object detection. Jia et al. [42] transfer cross-modal knowledge via visual-language permeation for few-shot learning. Wang et al. [43] incorporate generalized semantic knowledge into few-shot

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segmentation, improving generalization under limited labels. These studies collectively underscore the value of semantic guidance and cross-modal transfer, which align with and complement our prompt-based approach for HSI classification.

C. Prompt Learning

Prompt learning, as an efficient fine-tuning paradigm for natural language processing, has demonstrated significant advantages, particularly in applications of LLMs [21]. Its core mechanism lies in designing task-specific prompts to guide pretrained models to generate expected outputs, thereby effectively activating and utilizing the knowledge representations acquired during pretraining. In terms of implementation, prompt learning is mainly divided into two categories: hard prompts and soft prompts. Hard prompts typically refer to predesigned, fixed text templates or special token sequences that explicitly embed task instructions, input samples, and output formats into the input sequence in natural language form, requiring the model to make predictions based solely on this fixed structure [32]. In contrast, soft prompts are learnable parameter sequences obtained during model training, usually inserted into the input sequence as embedding vectors to adjust the model's internal representations or guide its attention, thereby facilitating the completion of downstream tasks [33]. With these advantages, prompt learning has also been extended to visual applications, where combining image features with designed prompts helps models better focus on key features related to classification tasks in images, thereby improving classification accuracy.

III. Methodology

A. Problem Definition

In the DPCFL framework, cross-domain knowledge transfer is realized through a domain adaptation strategy, where the source and target domains participate in synchronous training to achieve iterative model optimization. Throughout the whole training pipeline, the dual-prompt-driven module (DPDM) collaborates with the domain transfer learning process, integrating semantic guidance with visual features to alleviate prototype representation bias, strengthen cross-modal alignment, and ultimately enable few-shot HSI classification.

During the training process, the source domain image dataset is defined as D_s, and the target domain image dataset is defined as D_t. Here, D_s consists entirely of labeled samples, while D_t is composed of a small set of labeled samples D_l and a large set of unlabeled samples D_u. To facilitate the generation of class prototypes, the framework also incorporates label text sets T_s and T_t from different domains. The labeled sample sets collectively participate in the model's training optimization. In addition, the image dataset follows a few-shot learning setup with N-way-K-shot sampling. Given the limited number of samples in D_l, we employ Gaussian noise augmentation to expand the sample set to meet meta-training requirements. D_u is used to evaluate the model's performance. Furthermore, the combination of the image mapping layer and the image feature extractor (IFE) is defined as $f_\theta(\cdot)$, the text feature extractor as $g_\varphi(\cdot)$, and the adapter as $h_\phi(\cdot)$.

B. DPCFL Architecture

The overall architecture of DPCFL is shown in Fig. 2. This framework is improved based on prototype networks and mainly includes effective domain knowledge transfer and a dual-prompt-driven multimodal learning module. Specifically, the original data is first divided into 7×7 patches and sampled for training using meta-learning strategies with synchronous joint training of source and target domains. To address the spectral dimension differences between source and target domains, we use a mapping layer to align the spectral dimensions of data from both domains, eliminating feature space heterogeneity. The data are then input into a shared IFE, and source domain knowledge is effectively transferred through a cross-domain collaborative training mechanism. To effectively solve the prototype representation bias problem, a dual-prompt-driven learning module is introduced. This module integrates label semantic information with visual features to guide the generation of class-level prototypes. Different from conventional learnable class embeddings, text adapters, and CLIP-based prompt methods, it combines a static hyperspectral-specific prompt template and categorywise learnable dynamic prompt vectors optimized at the text feature level, together with multispace consistency constraints to mitigate semantic bias and prototype drift. On this basis, a joint metric space based on class-level prototypes and query set features is constructed to optimize the model's cross-domain representation capability.

During the testing phase, we adopt a KNN classifier instead of the standard nearest-prototype classifier. This choice is motivated by the cross-domain setting. The class-level prototype computed by averaging support samples in the target domain may be biased toward the source domain distribution, whereas KNN performs instance-level matching directly on the support set, which better adapts to the target domain's local data distribution. Importantly, KNN operates on the fixed feature representations extracted by the frozen encoder and mapping layer; thus, the learned image representations remain classifier-agnostic and can be readily used with alternative classifiers. In practice, we fit KNN using only five support samples per class (i.e., the same K-shot setting), which is sufficient to achieve stable performance, as verified by Pahde et al. [34] and our ablation study in Section IV-E.

C. Feature Extractor

1) Image Feature Extractor: The IFE, illustrated in Fig. 3, comprises a dual-branch network (DbN) and dynamic tanh (DyT) [35]. DbN extracts spatial and spectral features separately and combines them with equal weighting. The spatial branch uses 24 3-D convolution kernels of size $1 \times 1 \times 100$ for channel compression, followed by a residual block of two 2-D convolutional layers (3×3, 24) to enhance structural sensitivity and mitigate network degradation. A 7×7 average pooling integrates spatial information, with dropout for overfitting prevention, and a mapping layer (24, 128) projects the features. The spectral branch employs three convolutional layers with 24 kernels of $1 \times 1 \times 7$, with residual connections between the second and third layers, followed by a fourth layer

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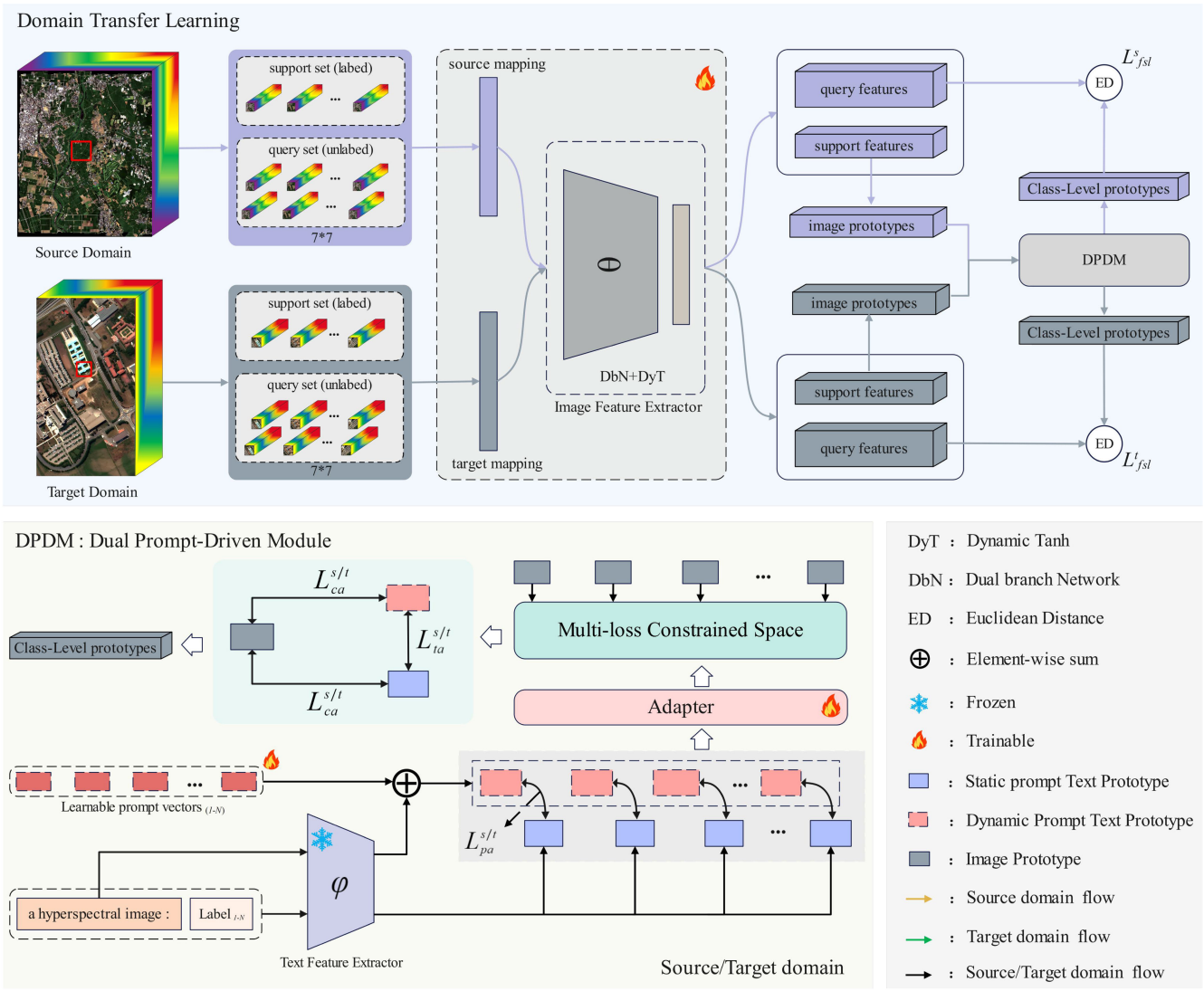


Fig. 2. Overall architecture of DPCFL. It is divided into domain transfer learning and the DPDM.

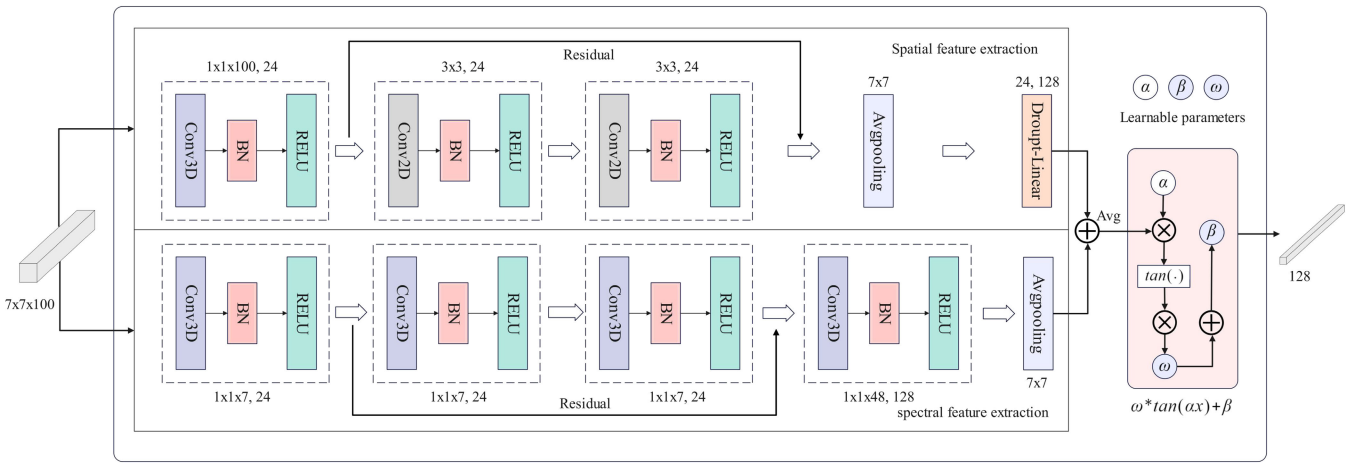


Fig. 3. Overview diagram of the IFE.

with 128 kernels of $1 \times 1 \times 48$ for feature transformation and dimensionality reduction to align with the spatial branch dimension. A 7×7 global average pooling yields a 128-D spectral vector. The spatial and spectral features are equally weighted to form a joint representation, which is then dynamically adjusted by the DyT mechanism [see (1)] to enhance

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feature discriminability and overall classification performance

$$\omega^* \tanh(\alpha x + \beta) \tag{1}$$

where x represents the joint spatial–spectral features, α serves as a global scalar parameter controlling nonlinear intensity, ω and β perform channel-adaptive scaling and shifting operations, respectively, and $\tanh(\cdot)$ is the hyperbolic tangent function that provides stable nonlinear mapping, thereby adaptively enhancing the discriminability of key features.

2) Text Feature Extractor: This architecture employs the CLIP-ViT-B/32 paired text encoder as the text feature extraction module. Based on a 12-layer Transformer architecture, this module encodes input label texts into 512-D feature vectors. Through large-scale contrastive learning training with image–text pairs, it has established effective semantic representation capabilities. The output features not only contain word-level semantic information but also capture deep contextual relationships. For downstream tasks, we keep the encoder parameters frozen to fully utilize its pretrained cross-modal representation characteristics. To better adapt to specific task requirements while maintaining the original encoder parameters unchanged, we introduce a lightweight adapter module. This module consists of two linear layers that transform text features into a 128-D target feature space through a combination of linear projection and ReLU activation. This design preserves both the powerful generalization capability of the pretrained model and provides necessary task adaptation capacity. It offers strong semantic prior support for subsequent spatial–spectral–textual multimodal feature fusion.

D. Domain Transfer Learning

To facilitate effective knowledge transfer from the source domain to the target domain, this study adopts a shared image encoder co-training strategy. The core approach involves using a parameter-shared encoder $f_\theta(\cdot)$ to map data from both domains into a unified embedding space for image feature extraction. In addition, category-matched episode sampling is performed on the source domain to simulate the target domain's class distribution. As shown in Fig. 4, few-shot learning tasks on both the source and target domains are executed synchronously throughout the whole training procedure. The shared encoder is iteratively optimized under the joint supervision of cross-domain data, enabling effective cross-domain knowledge transfer and adaptive feature alignment for domain adaptation.

Taking the source domain as an example, in each episode task, N categories are randomly selected, with the number matching the category count of the target domain. Each category contains K labeled support set samples and Q unlabeled query set samples. Based on this setup, the support set $S_s = \{(x_i^s, y_i^s)\}_{i=1}^{N\times K}$ and query set $Q_s = \{(x_j^s, y_j^s)\}_{j=1}^{N\times Q}$ are constructed. Subsequently, image prototypes are built by averaging the support set samples of each category, with details provided as follows:

$$p_c^s = \frac{1}{K} \sum_{x_i^s \in S_c^s} f_\theta\left(x_i^s\right) \tag{2}$$

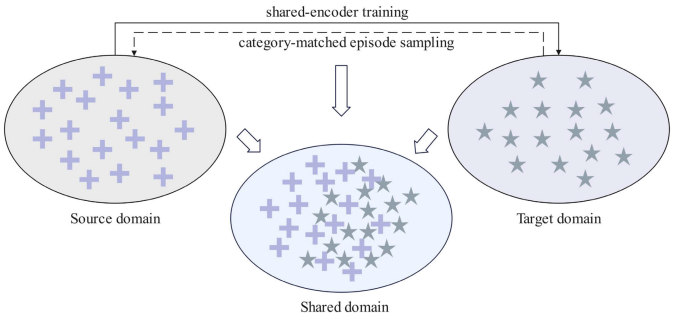


Fig. 4. Domain transfer learning. Shared encoder and category-matched episode sampling mitigate domain shift.

where p_c^s represents the image prototype for the cth category, and S_c^s denotes the collection of support set samples for the cth category in the source domain. The probability that query set sample x_j^s belongs to the cth category can then be expressed as follows:

$$P\left(y_j^s = c | x_j^s \in Q_s\right) = \frac{\exp\left(-ED\left(f_\theta\left(x_j^s\right), p_c^s\right)\right)}{\sum_{c=1}^{N} \exp\left(-ED\left(f_\theta\left(x_j^s\right), p_c^s\right)\right)} \tag{3}$$

where $ED(\cdot)$ denotes the Euclidean distance metric function, x_j^s represents the query set sample, and y_j^s is its corresponding label. Finally, the cross-entropy loss is adopted as the classification loss for the source domain, with details provided as follows:

$$L_{fsl}^s = -\sum_{\left(x_j^s, y_j^s\right) \in Q_s} log p\left(y_j^s = c | x_j^s\right). \tag{4}$$

The design of the target domain is consistent with that of the source domain and can be denoted as L_{fsl}^t. Therefore, the total loss can be expressed as follows:

$$L_{fsl} = L_{fsl}^s + L_{fsl}^t. \tag{5}$$

E. Dual-Prompt-Driven Learning

In the final stage of domain transfer learning, image prototypes are utilized to construct a metric space for optimizing model parameters. However, prototypes obtained solely by averaging image modality features fail to fully capture the essential characteristics of categories, resulting in insufficient prototype representation capability. To enhance prototype quality, we introduce textual modality information. A dual-prompt-driven learning strategy is adopted to jointly optimize dual-text representations composed of dynamic prompts and fixed prompts in both parameter space and feature space. Furthermore, in the cross-modal feature space, we employ the optimized dual-text representations as semantic guidance to iteratively refine the initial image prototypes, generating class-level prototypes that better approximate category centers. The evolution of prototypes in the feature space is specifically illustrated in Fig. 5.

The textual information of HSIs comes from a single source, primarily limited to label text names. To provide stable prior knowledge for the model, we designed a fixed prompt template: "a hyperspectral image: {label}." This template,

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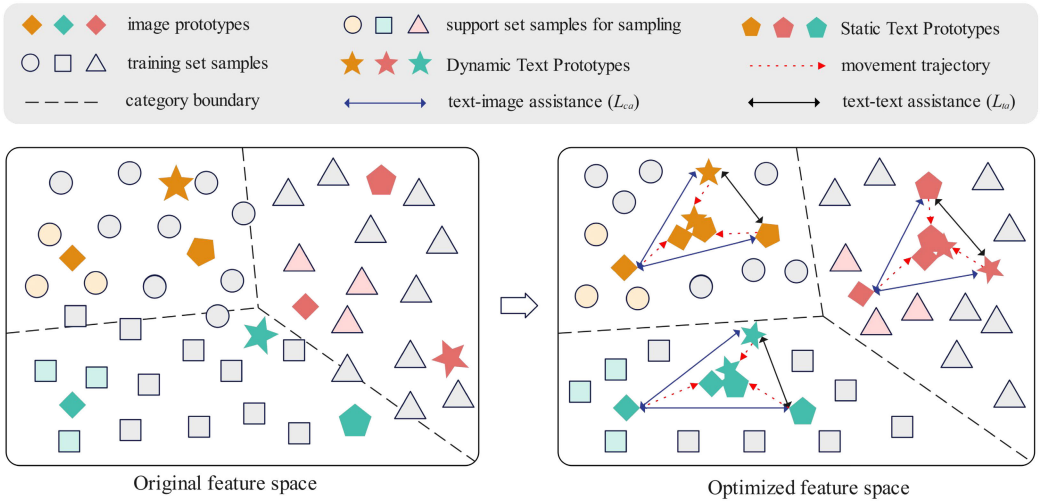


Fig. 5. Text-guided prototype refinement. Text prompts pull image prototypes toward category centers in the feature space, yielding sharper and more representative class-level representations.

combined with the label text, forms fixed textual inputs to construct the source domain and target domain text sets T_s and T_t. Due to the uniqueness of label texts, the acquisition of static text prototypes u_c^s directly relies on the text encoder $g_\varphi(\cdot)$ for feature extraction of the input t_c^s. Taking the source domain as an example, with details provided as follows:

$$u_c^s = g_\varphi\left(t_c^s\right). \tag{6}$$

However, the design of fixed prompt templates lacks flexibility and struggles to adapt to subtle semantic differences between different categories. To address this, we introduce learnable dynamic prompt vectors that combine with fixed prompts to form more adaptable dynamic text prototypes. Unlike conventional soft prompt methods, this approach no longer incorporates dynamic tokens processed by the CLIP model. This is because CLIP excels primarily at handling everyday semantic information but has limited capability in comprehending the more abstract semantics inherent in HSIs. By directly utilizing dynamic vectors, this method can more effectively capture the unique semantic characteristics of hyperspectral data. Specifically, for each category, a particular learnable dynamic prompt vector v_c^s is generated, with its dimensionality consistent with the output dimension of the encoder (512). Simultaneously, the fixed prompt "a hyperspectral image:" is separately fed into the text encoder to extract its corresponding fixed feature representation, denoted as vector v. The two are then elementwise summed to obtain the dynamic text prototype m_c^s, which can be expressed as follows:

$$m_c^s = v_c^s \oplus v. \tag{7}$$

The learnable dynamic prompt vector for each category serves as a category-specific semantic bias term. Unlike the fixed prompt template that provides a general context ("a hyperspectral image:"), v_c^s is optimized to encode fine-grained discriminative attributes, such as material composition, spectral absorption characteristics, and contextual land-cover properties, that are not explicitly expressed in the class name alone. By adding v_c^s to the fixed feature embedding v, the resulting dynamic text prototype adaptively emphasizes the unique visual-semantic signature of each category, while v anchors the representation to the stable HSI-related prior. This design ensures that the dynamic prompt complements rather than replaces the fixed prior, enabling flexible yet well-grounded category representation.

Based on the described operations, we construct a static text prototype set $U_s = \{u_1^s, u_2^s, \ldots, u_N^s\}$ and a dynamic text prototype set $M_s = \{m_1^s, m_1^s, \ldots, m_N^s\}$. Here, m_i^s carries inherent drift risks due to its dynamic nature, which may cause its representation to deviate from the true category semantics. Therefore, we impose a constraint mechanism in the parameter space, which can be expressed as follows:

$$L_{pa}^s = \frac{1}{N}\sum_{i=1}^{N}\left(m_i^s - u_i^s\right)^2. \tag{8}$$

The operations in the target domain follow the same approach and are denoted as L_{pa}^t. Consequently, the overall loss in the parameter space can be expressed as follows:

$$L_{pa} = L_{pa}^s + L_{pa}^t. \tag{9}$$

To further enhance the model's adaptability to downstream tasks, after applying the consistency constraints, both m_i^s and u_i^s are fed into the adapter module $h_\phi(\cdot)$ for joint optimization at the feature space level. This module improves the alignment of the two types of prototypes in feature representation, thereby strengthening their semantic consistency. To facilitate this process, we design a cosine similarity-based feature space alignment loss function, which can be expressed as follows:

$$L_{ta}^s = 1 - \frac{1}{N}\sum_{i=1}^{N}\cos\left(h_\phi\left(m_i^s\right), h_\phi\left(u_i^s\right)\right). \tag{10}$$

The implementation process in the target domain follows the same approach, denoted as L_{ta}^t. Thus, the overall text alignment loss in the feature space can be expressed as follows:

$$L_{ta} = L_{ta}^s + L_{ta}^t. \tag{11}$$

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Furthermore, the adapter also maps the text feature dimension to 128 dimensions to maintain consistency with the image feature dimension. After undergoing parameter space and feature space constraints, the dual-text prototypes are delivered to the multimodal feature space to assist in the generation of class-level prototypes. To better distinguish between pre and postoptimization states, we denote the optimized static text prototype set as $\hat{U}_s = \{\hat{u}_1^s, \hat{u}_2^s, \ldots, \hat{u}_N^s\}$ and the dynamic text prototype set as $\hat{M}_s = \{\hat{m}_1^s, \hat{m}_2^s, \ldots, \hat{m}_N^s\}$. In addition, we define the image–text set as $P_s = \{p_1^s, p_2^s, \ldots, p_N^s\}$. The set P_s forms two groups of positive and negative sample pairs with $\hat{U}_s$ and $\hat{M}_s$, respectively. Taking the group of P_s and $\hat{U}_s$ as an example, the pair $(p_i^s, \hat{u}_i^s)$ constitutes a positive sample pair, while all other combinations are treated as negative sample pairs. Subsequently, we design a loss function to achieve alignment in the cross-modal feature space, which can be expressed in (12), shown at the bottom of the page.

Thus, $D(p, u)$ denotes the negative Euclidean distance. Furthermore, to comprehensively account for different dominant perspectives, we design the loss function from two modal viewpoints. Consequently, the cross-modal loss for the combination of source domain P_s and $\hat{U}_s$ can be formulated as follows:

$$L_{ca}^s\left[P_s, \hat{U}_s\right] = \frac{1}{2N}\sum_{i=1}^{N}\left[L_{ca}^s\left(p_i^s, \hat{u}_i^s\right) + L_{ca}^s\left(\hat{u}_i^s, p_i^s\right)\right] \tag{13}$$

where N represents the batch size. The cross-modal learning on the combination of P_s and $\hat{M}_s$ follows the same approach, and its loss can be denoted as. Thus, the total cross-modal loss for the source domain can be expressed as follows:

$$L_{ca}^s = L_{ca}^s\left[P_s, \hat{U}_s\right] + L_{ca}^s\left[P_s, \hat{M}_s\right]. \tag{14}$$

The target domain follows the same design as the source domain, and its loss can be denoted as L_{ca}^t. Therefore, the total cross-modal alignment loss can be expressed by the following equation:

$$L_{ca} = L_{ca}^s + L_{ca}^t. \tag{15}$$

Finally, the total loss used by the model can be expressed as follows:

$$L_{total} = L_{fsl} + \lambda_1 L_{pa} + \lambda_2 L_{ta} + \lambda_3 L_{ca}. \tag{16}$$

In addition, to more clearly demonstrate the algorithmic process of DPCFL, we provide a detailed presentation of its training procedure in Algorithm 1.

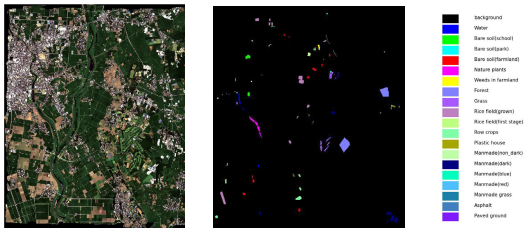
IV. Experiments

A. Datasets

The experiments involved a total of five datasets, including one source domain dataset and four target domain datasets. The source domain dataset is Chikusei (CH), containing 19 categories with diverse data characteristics that facilitate the model's learning of more generalizable representations. The target domain datasets consist of Indian Pines (IP), Salinas (SA), University of Pavia (UP), and Houston (HT), which are used to validate the effectiveness of the proposed method. Detailed information about the five datasets, including the number of spectral bands and spatial dimensions, is presented in Table I. The specific category information for each dataset is shown in Table II. In addition, Figs. 6–10 display the pseudocolor images, ground-truth maps, and category information for the different datasets.

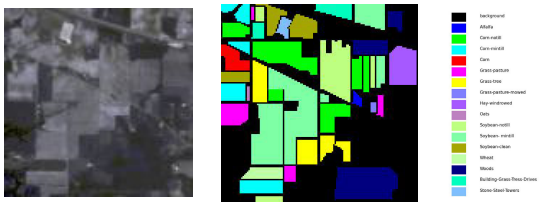
TABLE I
Basic Information of Five Datasets

Datasets	Source Domain	Target Domain			
	CH	IP	SA	UP	HT
Areas	Chikusei	Indiana	California	Pavia	Houston
Bands	128	102	204	103	144
Spatial size	2517 × 2335	145×145	512 × 217	610×340	349 × 1905
Spectral range	363–1018	400-2500	400-2500	430-860	380–1050



(a) (b) (c)

Fig. 6. Chikusei dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) Category of CH.



(a) (b) (c)

Fig. 7. IP dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) Category of IP.

B. Experimental Setups

The experiments were developed using the PyTorch framework with Python 3.9.18 as the runtime environment, configured with CUDA 11.3 for GPU acceleration. The hardware utilized an NVIDIA RTX 4090 GPU with 24-GB VRAM and 64-GB system memory.

$$L_{ca}^s\left(p_i^s, \hat{u}_i^s\right) = -log\frac{\exp\left(D\left(p_i^s, \hat{u}_i^s\right)/\tau\right)}{\sum_{k=1,k\neq i}^{N}\exp\left(D\left(p_i^s, \hat{u}_k^s\right)/\tau\right) + \sum_{k=1}^{N}\exp\left(D\left(p_i^s, \hat{u}_k^s\right)/\tau\right)} \tag{12}$$

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TABLE II
DETAILED CATEGORY INFORMATION OF SOURCE AND TARGET DOMAIN DATASETS—(CH–IP–SA–UP–HT)

Class	CH		IP		SA		UP		HT	
	Name	Count	Name	Count	Name	Count	Name	Count	Name	Count
1	Water	2345	Alfalfa	46	Brocoli green weeds 1	2009	Asphalt	6631	Healthy grass	1251
2	Bare soil (school)	2859	Corn notill	1428	Brocoli green weeds 2	3726	Meadows	18649	Stressed grass	1254
3	Bare soil (park)	236	Corn mintill	830	Fallow	1976	Gravel	2099	Synthetic grass	697
4	Bare soil (farmland)	48525	Corn	237	Fallow rough plow	1394	Tress	3064	Trees	1244
5	Natural plants	4297	Grass pasture	483	Fallow smooth	2678	Sheets	1345	Soil	1242
6	Weeds in farmland	1108	Grass trees in farmland	730	Stubble	3959	Bare soil	5029	Water	325
7	Forest	20516	Grass pasture mowed	28	Celery	3579	Bitumen	1330	Residential	1268
8	Grass	6515	Hay windrowed	478	Grapes untrained	11271	Bricks	3682	Commercial	1244
9	Rice field (groim)	13369	Oats	20	Soil vinyard develop	6203	Shadow	947	Road	1252
10	Rice field (first stage)	1268	Soybean notill	972	Corn senesced green weeds	3278			Highway	1227
11	Row crops	5961	Soybean mintill	2455	Lettuce romaine 4wk	1068			Railway	1235
12	Plastic house	2193	Soybean clean	593	Lettuce romaine 5wk	1927			Parking Lot 1	1233
13	Manmade (nan-dark)	1220	Wheat	205	Lettuce romaine 6wk	916			Parking Lot 2	469
14	Manmade (dark)	7664	Woods	1265	Lettuce romaine 7wk	1070			Tennis Court	428
15	Manmade (blue)	431	Buildings Grass Trees Drives	386	Vinyard untrained	7268			Running Track	660
16	Manmade (red)	222	Stone Steel Towers	93	Vinyard vertical trellis	1807				
17	Manmade grass	1040								
18	Asphalt	801								
19	Paved ground	145								
Total		77592		10249		54129		42776		15029

Algorithm 1 Training Process of DPCFL

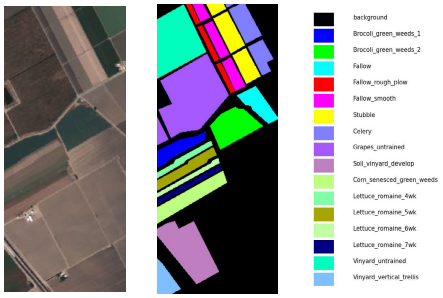
Input:

D_s: support set S_s, query set Q_s
D_l: support set S_t, query set Q_t
Text set: T_s and T_t

Output:

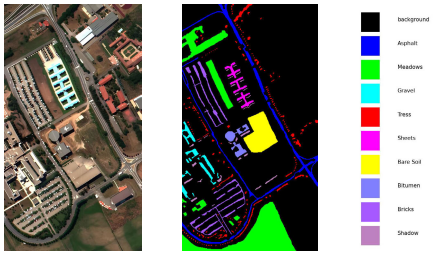
Construct prompt vector v_c^s for each category
for episode **do**
Calculate image prototype p_c^s and p_c^t by Eq. (2)
Static text prototype u_c^s and u_c^t by Eq. (6)
Dynamic text prototype m_c^s and m_c^t by Eq. (7)
for U_s and M_s **do**
Calculate L_{pa}^s and L_{ta}^s by Eq. (8) (10)
end for
Calculate L_{pa}^t and L_{ta}^t same as L_{pa}^s and L_{ta}^s
for $p_{T_i}^s \in P_T^s$ **do**
Calculate $L_{ca}^s[P_s, \hat{U}_s]$ by Eq. (12) (13)
end for
Calculate $L_{ca}^s[P_s, \hat{M}_s]$ same as $L_{ca}^s[P_s, \hat{U}_s]$
Calculate L_{ca}^s by Eq. (14)
for $(x_j^s, y_j^s) \in Q_s$ **do**
Calculate L_{fsl}^s by Eq. (3) (4)
end for
Calculate L_{fsl}^t same as L_{fsl}^s
Return total Loss L_{total} by Eq. (16)
Optimize the Model by L_{total}
end for

In the DPCFL method, HSIs were divided into $7 \times 7 \times B$ image patches as model inputs, where B represents the number of spectral bands. For the FSL tasks, the model used



(a) (b) (c)

Fig. 8. Salinas dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) Category of SA.



(a) (b) (c)

Fig. 9. UP dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) Category of UP.

the Adam optimizer with a learning rate of 0.0001 and was trained for 5000 episodes. The meta-tasks followed an N-way-K-shot-Q-query sampling strategy, where N was set to the number of target domain categories (16 for IP and SA, 9 for UP, and 15 for HT). K = 1 and Q = 19, meaning each episode contained 1 support sample and 19 query samples per class. The support set was drawn from a Gaussian-noise-augmented training set constructed from five labeled

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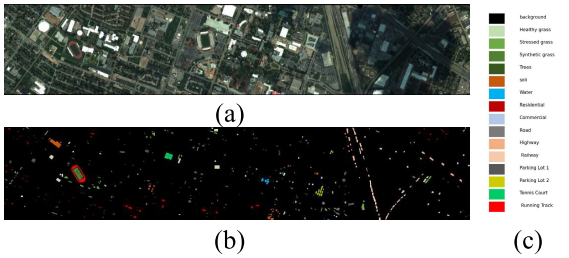


Fig. 10. Houston dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) Category of HT.

TABLE III
PARAMETER TUNING OF λ_1 IN L_{pa} (OA AS METRIC)

Datasets	λ_1				
	0.5	1.0	1.5	2.0	2.5
IP	81.20	**81.68**	81.39	81.49	81.23
SA	93.12	93.45	**93.46**	93.38	93.21
UP	84.89	85.09	84.91	**85.20**	84.93
HT	79.52	**79.58**	79.46	79.55	79.32

samples per class. To objectively evaluate model performance, three standard metrics were adopted: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa). All experiments were repeated over ten independent runs with different random seeds, and the results are reported as mean ± standard deviation to ensure a stable and reliable assessment.

C. Parameter Tuning

In the optimization process of the model, a total of four loss calculations are involved. Among them, the weight of the loss L_{fsl} is fixed at 1, serving as the base weight. The weights of the other three loss terms are set as adjustable parameters, aiming to flexibly adjust their contribution to the total loss according to specific task requirements. To systematically explore the optimal combination of loss weights, an iterative selection strategy was adopted in the experiment, which involved adjusting the parameters of each adjustable weight within the predefined candidate set {0.5, 1.0, 1.5, 2.0, 2.5} to achieve better model performance. The specific experiments are shown in Tables III–V. We first obtain the optimal result for λ_1 and then fix this parameter value, followed by adjusting λ_2 parameters to continue the experiment. The same strategy is applied for λ_3. The optimal results for IP, UP, SA, and HT are achieved at $\{\lambda_1, \lambda_2, \lambda_3\} = \{1.0, 1.0, 1.0\}, \{2.0, 1.0, 1.0\}, \{1.5, 1.0, 1.5\}$, and $\{1.0, 1.0, 1.5\}$, respectively.

As shown in Table IV, the weight λ_2 for textual feature alignment exhibits exceptional stability across all datasets, with its optimal value consistently converging to 1.0 and fluctuations confined within 0.5%. This is because λ_2 merely performs linear scaling in the feature space via cosine similarity loss without altering the topological structure of class separability. Consequently, λ_2 was fixed at 1.0 in the parameter interaction experiments to streamline the optimization process. Fig. 11 further reveals the tradeoff between λ_1 and λ_3. The experimental results indicate that a moderate λ_1 provides a

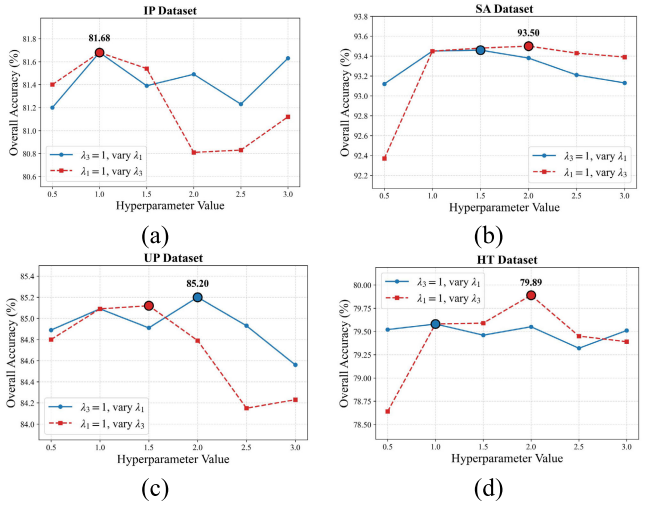


Fig. 11. Comparative results of varying $\lambda 1$ and $\lambda 3$ on IP, SA, UP, and HT datasets. (a) IP dataset. (b) SA dataset. (c) UP dataset. (d) HT dataset.

TABLE IV
PARAMETER TUNING OF λ_2 IN L_{ta} (OA AS METRIC)

Datasets	λ_2				
	0.5	1.0	1.5	2.0	2.5
IP	81.52	**81.68**	81.25	81.51	81.49
SA	93.25	**93.46**	93.32	93.28	93.27
UP	84.58	**85.20**	84.64	84.59	84.65
HT	79.45	**79.58**	79.49	79.47	79.43

stable semantic reference for cross-modal alignment, whereas an excessively high λ_1 does not restrict the model's representational flexibility but leads to overly conservative optimization, resulting in a slight performance decline. When λ_1 is maintained around 1.0, λ_3 effectively enhances performance within the range of 1.0–1.5. However, if λ_1 is too high, the model tends to over-preserve the original state of textual features, thereby weakening its responsiveness to cross-modal discrepancies and limiting further performance gains. Furthermore, with λ_1 fixed at 1.0, the ample parameter redundancy provided by the adapter structure ensures that moderately increasing λ_3 does not negatively impact model flexibility. Instead, it maximizes classification performance while guaranteeing training robustness.

D. Comparative Experiment

To evaluate the effectiveness of the proposed method DPCFL under sample-limited conditions, this study compares it with eight different benchmark methods. These eight methods are SVM [3], 3DCNN [6], SSRN [7], DCFSL [16], Gia-CFSL [36], ADAFSL [37], MLRL [38], and FDFSL [39]. The specific experimental details for the four target domain datasets (IP, UP, SA, and HT) can be found in Tables VI–IX. From Table VI, it can be observed that the results of the DPCFL method on the IP dataset in terms of OA, AA, and Kappa are 81.68%, 89.02%, and 79.30%, respectively, with an improvement of at least 11.48% in the main metric

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TABLE V
PARAMETER TUNING OF λ_3 IN L_{ca} (OA AS METRIC)

Datasets	λ_3				
	0.5	1.0	1.5	2.0	2.5
IP	81.40	**81.68**	81.54	80.81	80.83
SA	93.20	93.46	**93.47**	93.26	93.12
UP	84.58	**85.20**	84.77	85.18	85.15
HT	79.25	79.58	**79.64**	79.59	79.51

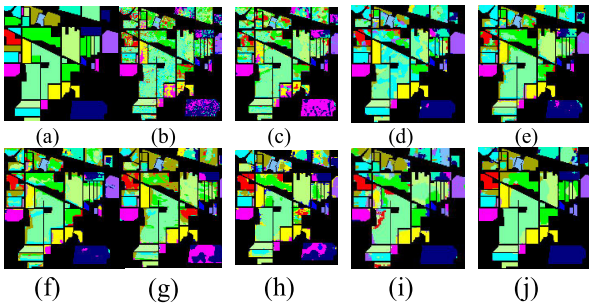


Fig. 12. Comparison of classification effects on the IP dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) Gia-CFSL. (g) ADAFSL. (h) MLRL. (i) FDFSL. (j) Ours.

OA compared to other methods. From Table VII, it can be observed that the results of the DPCFL method on the SA dataset in terms of OA, AA, and Kappa are 93.50%, 96.50%, and 92.76%, respectively, with an improvement of at least 2.63% in the main metric OA compared to other methods. From Table VIII, it can be observed that the results of the DPCFL method on the UP dataset in terms of OA, AA, and Kappa are 85.20%, 91.06%, and 81.22%, respectively, with an improvement of at least 1.89% in the main metric OA compared to other methods. From Table IX, it can be observed that the results of the DPCFL method on the HT dataset in terms of OA, AA, and Kappa are 79.89%, 82.69%, and 78.03%, respectively, with an improvement of at least 1.07% in the main metric OA compared to other methods. Overall, our method achieved optimal performance across three different metrics, further demonstrating that the introduction of textual information can effectively capture potential semantic information, thereby enhancing the model's comprehension and predictive accuracy. To provide an intuitive comparison of classification performance across different methods, we present visual results in Figs. 12–15. It can be observed that, for categories with highly similar spectral characteristics, the proposed method demonstrates comparatively better discriminative capability. This is particularly evident in Fig. 12 for the IP dataset. Among the classes "Corn-notill," "Corn-min till," and "Corn," the proposed approach yields fewer misclassified pixels and preserves clearer class boundaries. In addition, we present a visual comparison of our proposed method with state-of-the-art (SOTA) methods on t-SNE plots in Fig. 16, where it can be observed that the DPCFL method exhibits the best distribution effects across the four datasets.

In addition, to further demonstrate the effectiveness of the dual-prompt-driven method design, we compared the

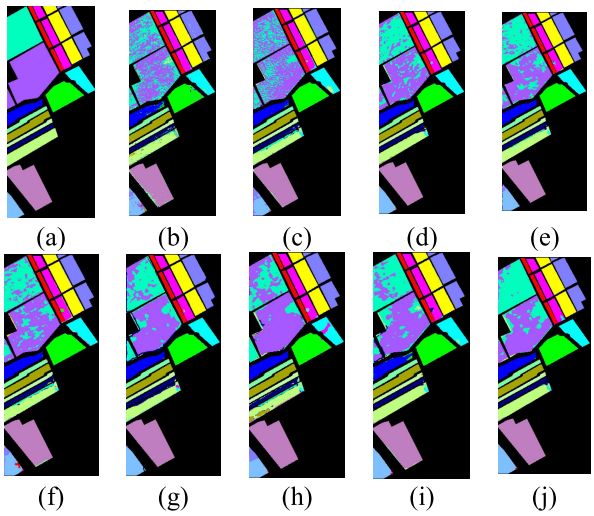


Fig. 13. Comparison of classification effects on the SA dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) Gia-CFSL. (g) ADAFSL. (h) MLRL. (i) FDFSL. (j) Ours.

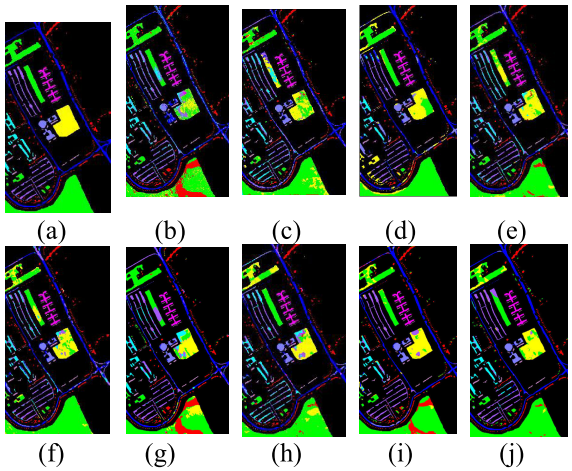


Fig. 14. Comparison of classification effects on the UP dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) Gia-CFSL. (g) ADAFSL. (h) MLRL. (i) FDFSL. (j) Ours.

performance of three different approaches: no prompting, hard prompting, and soft prompting. In the hard prompting group, we compared our designed template "a hyperspectral image: {label}" with three types of hard prompts: generic feature descriptions ("a photograph of {label}"), remote sensing-specific descriptions, and retrieval-augmented generation (RAG)-enhanced descriptions. In the soft prompting group, we compared our designed soft prompting strategy "Vhard + [Vlabel(1-N)]" with the existing soft prompting design "[V1][V2]... [VM] {label}." The specific experimental results are detailed in Table X, where all assessments are based on the OA metric. The analysis indicates that, compared to existing methods, both hard and soft prompting under the DPCFL framework exhibit superior performance. The combination of the two prompting strategies further demonstrates the best overall effect, which also confirms the effectiveness of the DPCFL method.

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TABLE VI
COMPARISON RESULTS (%) BASED ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	Ours
1	81.95	85.04	99.01	97.80	93.41	97.80	98.29	95.37	**99.27**
2	32.48	31.39	42.91	45.14	47.98	52.18	48.73	53.89	**71.99**
3	36.41	30.22	58.84	46.80	49.35	60.16	62.33	64.08	**70.88**
4	56.59	37.82	81.49	75.60	76.51	86.77	88.62	83.02	**96.38**
5	67.01	62.50	78.63	67.74	66.65	78.70	78.56	74.12	**80.52**
6	65.39	62.91	95.75	86.50	84.14	86.03	88.99	87.24	**97.48**
7	91.74	92.23	99.55	99.57	99.57	99.57	99.57	99.57	**100**
8	65.35	59.86	76.90	76.03	83.76	85.31	82.03	86.77	**96.03**
9	80.00	95.24	**100**	**100**	**100**	**100**	**100**	**100**	**100**
10	36.71	50.78	60.51	57.08	58.23	69.71	67.81	63.02	**81.51**
11	41.56	51.70	51.14	61.45	61.32	58.66	61.37	63.83	**74.44**
12	27.93	27.27	62.59	48.69	45.83	**81.04**	52.82	54.69	79.00
13	86.55	91.34	**99.29**	95.35	96.45	97.65	98.90	98.95	99.05
14	62.96	59.24	87.98	89.48	87.83	84.10	88.44	87.75	**91.52**
15	29.19	46.15	75.21	67.72	70.21	79.50	74.91	76.48	**87.69**
16	89.63	81.08	97.70	96.82	96.0	98.30	**99.77**	97.95	98.64
OA	46.77 ±2.94	49.02 ±4.18	65.72 ±3.77	64.89 ±3.50	65.44 ±2.23	68.61 ±3.54	69.38 ±3.87	70.20 ±2.66	**81.68 ±1.90**
AA	59.45 ±1.93	60.30 ±2.30	79.22 ±1.61	75.74 ±2.44	76.08 ±1.48	80.34 ±1.96	80.70 ±1.22	80.42 ±1.95	**89.02 ±1.40**
Kappa	40.5 5±2.95	42.63 ±4.28	61.72 ±3.90	60.36 ±3.96	61.02 ±2.53	64.76 ±3.78	65.70 ±4.11	66.48 ±2.94	**79.30 ±2.11**

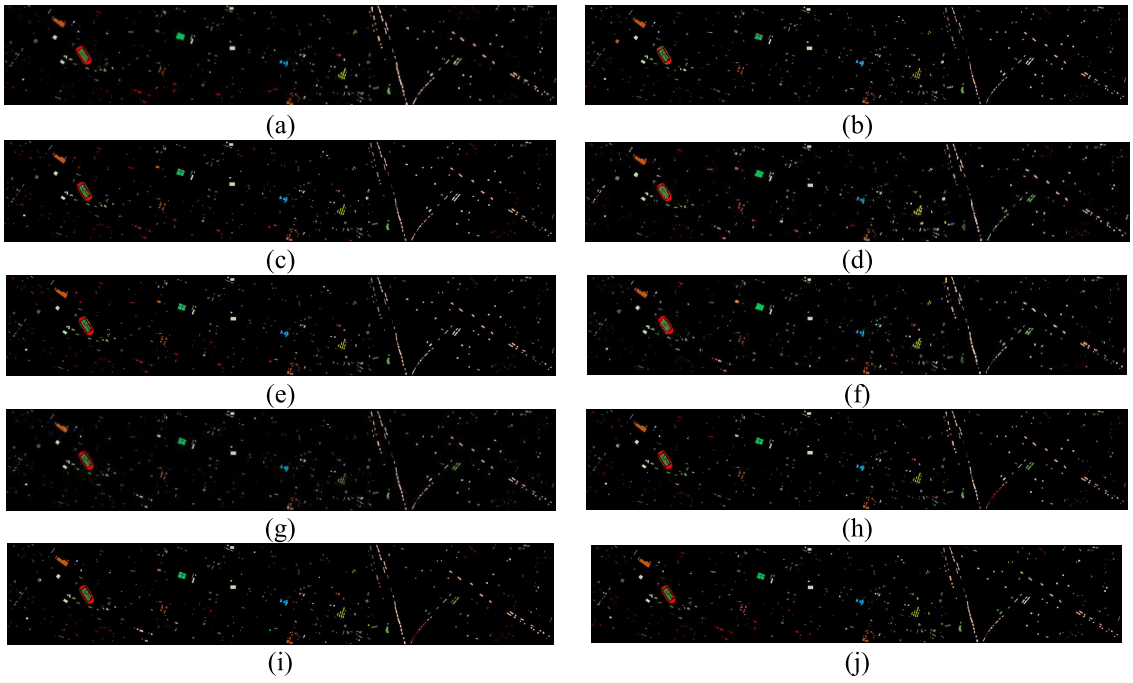


Fig. 15. Comparison of classification effects on the HT dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) Gia-CFSL. (g) ADAFSL. (h) MLRL. (i) FDFSL. (j) Ours.

E. Ablation Study

To verify the effectiveness of the components of the DPCFL method, we conducted ablation experiments. As shown in Table XI, this experiment decomposes DPCFL into two core components: the IFE and the DPDM. The IFE consists of a DbN and DyT. The DPDM includes both static prompts and dynamic prompt vectors. It is worth noting that when the static

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TABLE VII
COMPARISON RESULTS (%) BASED ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	Ours
1	96.02	93.08	98.53	99.45	98.09	99.63	**99.97**	98.86	99.49
2	94.71	95.99	99.67	98.82	99.40	99.96	98.75	99.27	**100**
3	88.56	89.80	96.87	97.31	95.46	**97.75**	88.73	91.21	97.69
4	98.83	98.15	99.31	99.26	98.52	**99.83**	99.21	98.91	98.40
5	94.01	92.34	95.79	91.52	86.55	96.20	91.12	94.13	**97.19**
6	98.39	98.80	99.66	99.84	99.25	99.84	99.59	98.97	**99.99**
7	96.61	97.83	99.93	99.69	99.28	98.75	99.59	99.08	**99.97**
8	56.11	59.31	66.17	79.48	81.31	77.51	80.63	79.37	**83.42**
9	94.04	96.93	99.79	99.32	97.76	99.52	99.69	98.59	**99.88**
10	76.44	78.10	90.73	86.78	83.73	85.93	85.83	85.50	**92.06**
11	91.06	89.85	97.34	97.01	93.56	**99.72**	97.66	96.39	99.07
12	99.29	99.34	**99.88**	99.17	99.19	99.83	98.73	99.39	99.96
13	95.51	95.17	99.01	97.85	94.27	**99.49**	98.49	97.67	99.01
14	91.87	91.24	98.13	98.43	98.46	98.56	98.15	**99.52**	97.19
15	64.62	59.06	81.47	75.73	74.51	75.05	72.62	81.23	**84.87**
16	84.08	83.21	94.43	92.75	90.32	97.21	92.33	91.54	**95.77**
OA	81.42 ±2.99	81.74 ±2.31	89.13 ±2.59	90.51 ±0.99	89.75 ±0.80	90.52 ±1.90	89.97 ±1.73	90.85 ±0.90	**93.50 ±0.51**
AA	88.76 ±1.34	88.64 ±1.57	94.79 ±0.55	94.52 ±0.66	93.10 ±1.19	95.30 ±0.99	93.82 ±1.54	94.35 ±0.98	**96.50 ±0.74**
Kappa	79.44 ±3.23	79.72 ±2.53	87.95 ±2.80	89.45 ±1.09	88.60 ±0.90	89.48 ±2.09	88.83 ±1.92	89.83 ±0.99	**92.76 ±0.56**

TABLE VIII
COMPARISON RESULTS (%) BASED ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	Ours
1	88.98	59.82	**91.84**	78.09	73.25	80.56	77.65	86.07	88.82
2	83.91	63.05	**95.13**	81.63	84.95	81.72	81.39	83.09	78.07
3	39.98	68.91	55.23	64.72	62.70	72.09	61.55	65.16	**82.27**
4	60.22	77.31	78.02	**94.65**	94.05	87.90	87.81	89.29	92.04
5	95.44	90.77	98.34	99.38	99.65	98.45	99.81	99.40	**99.94**
6	37.12	63.40	53.56	74.94	73.73	73.70	69.49	75.11	**89.70**
7	40.62	87.64	60.07	78.44	81.94	87.05	82.48	88.66	**98.69**
8	68.17	57.27	85.34	68.25	72.34	76.84	70.15	85.31	**90.89**
9	99.13	95.57	98.08	98.62	96.18	99.11	96.88	94.62	99.09
OA	64.12 ±4.55	65.74 ±1.77	76.26 ±5.78	80.08 ±3.92	80.91 ±1.29	81.22 ±4.52	78.88 ±3.65	83.31 ±1.89	**85.20 ±3.58**
AA	68.18 ±2.27	73.72 ±1.01	79.51 ±3.21	82.08 ±1.91	82.09 ±2.04	84.16 ±2.57	80.80 ±2.11	85.19 ±2.23	**91.06 ±1.35**
Kappa	55.59 ±4.64	57.37 ±1.97	70.56 ±6.69	74.46 ±4.45	75.33 ±1.53	75.82 ±5.15	72.80 ±4.19	78.84 ±2.16	**81.22 ±4.18**

prompts operate alone, their mechanism only involves the interaction between the loss function L_{ca} and image features. Similarly, when the dynamic prompt vectors operate alone, their functionality is also limited to the application of L_{ca}.

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TABLE IX
COMPARISON RESULTS (%) BASED ON THE HT DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	Ours
1	81.03	89.24	81.94	**90.20**	89.66	89.56	88.68	87.38	88.19
2	74.54	77.15	80.72	74.45	74.43	87.56	83.75	88.06	**88.26**
3	97.60	95.06	99.68	94.65	93.48	97.33	96.40	97.90	**99.60**
4	86.28	88.15	89.81	91.38	89.18	89.96	92.12	92.07	**93.16**
5	89.06	94.68	95.78	96.90	94.69	95.73	94.77	**99.42**	99.15
6	82.52	72.12	83.57	84.34	79.61	78.34	82.44	85.06	**88.44**
7	55.96	54.53	66.79	59.29	58.38	60.84	73.50	69.70	**75.33**
8	24.85	42.38	22.36	42.59	37.74	42.24	42.18	**48.42**	39.79
9	58.49	59.98	61.50	59.95	55.93	56.12	68.91	**73.17**	68.27
10	47.67	53.29	49.14	66.79	60.37	**68.54**	62.63	64.97	69.12
11	51.04	61.13	**77.97**	56.02	59.46	65.55	60.33	64.52	76.11
12	39.67	44.77	40.70	**75.42**	71.84	71.87	65.18	62.14	66.14
13	19.61	31.56	75.09	86.73	88.64	44.94	91.25	**93.00**	92.37
14	93.48	86.36	**99.80**	88.60	87.42	94.52	89.67	98.16	99.55
15	96.05	96.56	99.77	94.65	91.41	94.99	97.05	99.39	**100**
OA	63.75 ±2.26	68.58 ±2.58	70.46 ±1.24	74.22 ±2.92	72.26 ±3.60	74.77 ±2.85	76.63 ±2.28	78.57 ±1.98	**79.89 ±2.18**
AA	66.52 ±1.91	69.80 ±2.61	74.97 ±0.99	77.47 ±2.62	75.48 ±2.82	75.87 ±3.01	79.30 ±2.10	81.56 ±1.55	**82.90 ±1.74**
Kappa	60.89 ±2.39	65.92 ±2.76	68.08 ±1.33	72.16 ±3.16	70.05 ±3.89	72.72 ±3.08	74.75 ±2.47	76.86 ±2.12	**78.30 ±2.35**

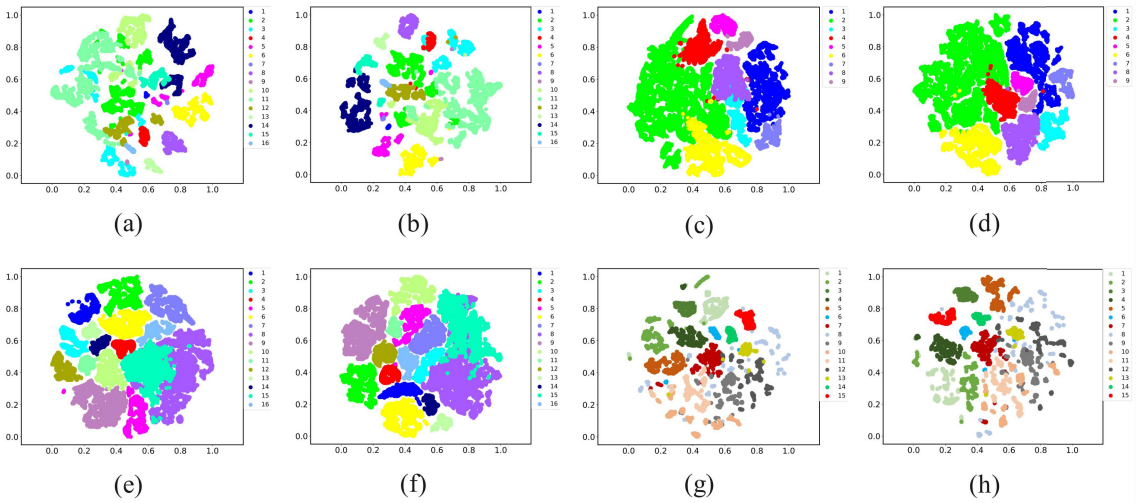


Fig. 16. T-SNE distribution analysis compared with the SOTA method. (a) FDFSL on IP. (b) Ours on IP. (c) FDFSL on UP. (d) Ours on UP. (e) FDFSL on SA. (f) Ours on SA. (g) FDFSL on HT. (h) Ours on HT.

The ablation results presented in Table XI elucidate the distinct contributions of each component within the proposed framework. The IFE, composed of the DbN and DyT, establishes a robust spectral–spatial baseline, while the DyT module further enhances discriminability through adaptive feature modulation that is particularly critical for cross-domain scenarios. The static prompt, even when used alone with the cross-modal loss L_{ca}, provides stable prior guidance and improves OA by approximately 0.6%–1.2% over the IFE-only baseline across all datasets, confirming the benefit of textual priors for prototype refinement. The dynamic prompt alone offers slightly better category-specific adaptation than the static prompt. When both prompts are combined without the parameter-space and feature-space constraints, the gain is modest, reaching only 81.18% on IP, which indicates that simple fusion is insufficient to fully exploit their complementary strengths. Introducing the

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TABLE X
PERFORMANCE COMPARISON OF DIFFERENT PROMPTING METHODS (OA IS USED AS THE EVALUATION METRIC)

Promp Learning		IP	SA	UP	HT
No Prompt	{label}	79.63	92.41	83.34	79.12
Hard Prompt	a photo of {label}[18]	79.51	92.58	83.56	79.05
	An aerial image includes some {label}[40]	79.94	92.68	83.64	79.18
	a photo of {label}: feature description	79.74	92.48	83.43	78.53
	RAG-enhanced descriptions	78.87	91.98	82.53	78.10
	a hyperspectral image : {label} (ours)	<u>80.06</u>	<u>92.76</u>	<u>83.67</u>	<u>79.28</u>
Soft Prompt	$[V_1][V_2]\ldots[V_M]$ {label} [22]	80.18	<u>92.87</u>	83.72	79.30
	$V_{\text{hard}} + [V_{\text{label(1-N)}}]$ (ours)	<u>80.24</u>	92.86	<u>83.98</u>	<u>79.35</u>
Dual Prompt	DPCFL (ours)	**81.68**	**93.50**	**85.20**	**79.89**

TABLE XI
ABLATION EXPERIMENT IN DIFFERENT MODULES (FIVE LABELED SAMPLES PER CLASS). OA IS USED AS THE EVALUATION METRIC, WHERE UNDERLINED REPRESENTS THE SECOND-BEST AND BOLD REPRESENTS THE BEST

IFE		DPDM				IP	SA	UP	HT
DbN	DyT	Static Prompt	Dynamic Prompt Vectors						
		L_{ca}	L_{ca}	L_{pa}	L_{ta}				
√						75.43	91.20	82.86	78.20
√	√					79.46	92.38	83.27	79.18
√		√				80.06	92.76	83.67	79.28
√			√			79.98	92.78	83.79	79.29
√	√	√				80.12	92.81	83.82	79.31
√	√		√			80.24	92.86	83.98	79.35
√	√	√	√			81.18	93.14	84.32	79.49
√	√	√	√	√		<u>81.25</u>	93.19	84.57	<u>79.58</u>
√	√	√	√		√	81.20	<u>93.27</u>	<u>84.96</u>	79.54
√	√	√	√	√	√	**81.68**	**93.50**	**85.20**	**79.89**

TABLE XII
RESULTS OF DIFFERENT CLASSIFICATION HEADS

	IP	SA	UP	HT
Proto-Metric	80.12	92.58	84.67	79.03
KNN	**81.68**	**93.50**	**85.20**	**79.89**

TABLE XIII
PERFORMANCE COMPARISON OF DyT AND STANDARD ACTIVATIONS

	IP	SA	UP	HT
DyT	**81.68**	**93.50**	**85.20**	**79.89**
GELU	78.16	91.37	81.52	76.94
ReLU	80.34	92.15	83.72	78.23

parameter-space constraint L_{pa} prevents excessive deviation of dynamic prompt vectors from the static anchor. In contrast, the feature-space constraint L_{ta}, which enforces cosine similarity between static and dynamic text prototypes in the Adapter feature space, yields a more notable gain. Ultimately, the full DPCFL model, which integrates all three constraints, achieves the best performance. This hierarchy of constraints, where L_{pa} prevents catastrophic drift, L_{ta} ensures feature-level coherence, and L_{ca} directly aligns text prototypes with image prototypes, jointly enables stable and accurate semantic alignment.

We further conduct ablation experiments to verify the rationality of using the KNN classifier. Under identical experimental settings, we compare KNN with the standard nearest-prototype classifier. As shown in Table XII, the nearest-prototype classifier yields relatively lower OA than the KNN counterpart. This performance gap mainly stems from the inherent cross-domain spectral shift between the

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TABLE XIV
COMPUTATIONAL COMPLEXITY ANALYSIS WITH DIFFERENT METHODS

Dataset	Metrics	Method								
		SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	Ours
IP	Training time	2.58s	112.78s	159.64s	2101.45s	4269.32s	2501.45s	3633.43	2453.21s	911.25s
	Testing time	0.25s	4.26s	13.56s	5.67s	2.56s	6.53s	1.73S	5.79s	0.55s
	#Params	--	0.07M	0.35M	4.28M	0.86M	3.21M	6.08M	5.28M	0.36M
UP	Training time	0.31s	53.17s	67.52s	1534.98s	4001.86s	2980s	2843.12s	2857s	504.12s
	Testing time	0.65s	6.25s	31.43s	6.84s	5.33s	19.23s	5.50s	15.43s	1.91s
	#Params	-	0.10M	0.26M	4.25M	0.85M	3.19M	6.07M	5.25M	0.35M
SA	Training time	3.46s	152.36s	219.34s	2760.51s	4252.35s	3120.12s	3636.96s	3136.15	851.37s
	Testing time	1.25s	15.27s	52.17s	23.54s	10.25s	21.25s	8.97s	26.78s	3.05s
	#Params	--	0.07M	0.35M	4.28M	0.86M	3.21M	6.08M	5.28M	0.36M
HT	Training time	1.93s	99.47s	113.56s	1856.23s	3253.28s	2563.4s	2438.25s	1832.52s	749.31s
	Testing time	0.12s	4.12s	10.25s	4.68s	1.98s	5.17s	3.65s	4.32s	0.68s
	#Params	--	0.07M	0.26M	4.28M	0.85M	3.19M	6.07M	5.25M	0.35M

source and target domains. By contrast, KNN performs fine-grained matching on support samples and better adapts to the target-domain distribution, yielding more stable performance in cross-domain few-shot scenarios.

We further investigate the efficacy of DyT embedded within the IFE for dynamic modulation of joint spectral–spatial features in cross-domain few-shot hyperspectral classification. Comparative results against two prevailing static activation functions, namely ReLU and GELU, are summarized in Table XIII to reveal the underlying working principle of DyT. From a functional perspective, DyT jointly mitigates cross-domain spectral variability and implements adaptive feature scaling. Traditional static activation functions adopt fixed mapping rules across all feature samples, where ReLU is prone to neuron deactivation and consequent loss of critical spectral–spatial information, and GELU yields over-smoothed feature representations that degrade cross-domain feature discriminability. By contrast, DyT employs dynamic nonlinear transformation to adaptively calibrate feature responses conditioned on input spectral–spatial attributes, which effectively alleviates spectral distribution divergence between source and target domains while preserving fine-grained discriminative cues. This adaptive modulation mechanism enables the acquisition of more robust feature representations under scarce labeled samples, thereby validating the indispensability of DyT for cross-domain few-shot HSI classification.

F. Computational Complexity Analysis

To systematically evaluate the differences in computational efficiency among various methods, this study presents the training time, testing time, and model parameter count for each method in Table XIV. These metrics are key dimensions for quantifying the computational complexity of algorithms and can objectively reflect the performance of different methods in terms of resource consumption. From the table, it can be observed that although the DPCFL method introduces semantic information from the textual modality, it does not lead to excessive resource waste compared to SOTA methods; instead, it achieves a certain degree of optimization in computational efficiency. While DPCFL may consume more resources compared to traditional methods (such as SVM), this investment results in significant performance improvements. In other words, the additional computational cost incurred is justified. Overall, the DPCFL method not only effectively reduces the consumption of computational resources but also enhances performance.

V. CONCLUSION

This article proposes a DPCFL method to address the semantic bias issue in small-sample HSI classification. By incorporating the textual modality, the model can better capture latent semantic information, thereby enhancing its ability to distinguish different land cover classes. However, the inherent semantic similarity in textual information may introduce noise and interfere with the discrimination process. To this end, this study integrates the strengths of hard and soft prompts to optimize textual representations from three aspects: parameter space, feature space, and cross-modal space. This optimization improves the quality and discriminability of textual representations, which in turn supports the generation of more discriminative class prototypes and significantly enhances the model's classification performance for similar classes. Experiments on four public hyperspectral datasets (IP, UP, SA, and HT) demonstrate that DPCFL achieves optimal classification results under small-sample settings. The primary contribution of this research is the proposal of a cross-modal learning framework that combines dual prompt mechanisms and multispace optimization, effectively mitigating semantic bias in small-sample scenarios and providing new insights for feature representation learning in HSIs. However, the current method still has certain limitations. The interaction mechanism between the image and textual modalities is not tight enough, and prompt learning focuses solely on optimizing the textual modality while neglecting in-depth exploration of the image-side prompt mechanism. Therefore, in future research, we will further explore the potential of prompt learning in the image modality, focusing on the deep interaction mechanism between textual and image prompts to build a more unified and robust

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multimodal learning framework and continuously improve the model's discrimination and generalization capabilities.

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Yuhang Li received the B.E. degree in software engineering from the School of Computer Science, Baoji University of Arts and Sciences, Baoji, China, in 2022. He is currently pursuing the master's degree in electronic information with the School of Mathematics and Computer Science, Yan'an University, Yan'an, China.

His research topics mainly include: computer vision, few-shot learning, and hyperspectral image analysis.

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Jinrong He (Member, IEEE) received the B.S. degree in science of information and computation, and the M.S. degree in computational mathematics from Wuhan University of Technology, Wuhan, China, in 2007 and 2010, respectively, and the Ph.D. degree in computer software and theory from the State Key Laboratory of Software Engineering, Wuhan University, Wuhan, in 2014.

From 2015 to 2018, he was a Lecturer at Northwest A&F University, Xianyang, China. He is currently a Professor at Yan'an University, Yan'an. His research interests include computer vision, machine learning, and few-shot learning.



Xiangqing Zhang received the Ph.D. degree from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2024.

He is currently a Lecturer with the College of Mathematics and Computer Science, Yan'an University, Yan'an, China. His research interests include deep learning, data mining, and object detection of remote sensing images.



Zhaokui Li (Member, IEEE) received the M.S. degree in computer application from Liaoning University, Shenyang, China, in 2003, and the Ph.D. degree in computer software and theory from Wuhan University, Wuhan, China, in 2014.

He is currently a Professor with the School of Computer, Shenyang Aerospace University, Shenyang. His research interests include hyperspectral image analysis, computer vision, and machine learning.

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